SHORT EGYPTIAN FRACTIONS

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Abstract. We prove that every fraction $\frac{a}{b} \in (0,1)$ can be written as

$$\frac{a}{b} = \frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_r}$$

where the x_i are distinct positive integers and

$$r \leq \left(\frac{16}{\log 2} + o(1)\right) \log\log b.$$

This resolves an old conjecture of Erdős [1, 3, 4] and is optimal up to the multiplicative constant. This paper gives a human-written exposition and simplification of a proof that was recently posted by OpenAI [6].

1. Adapting the old proofs

We are interested in writing fractions as short sums of unit fractions. The classical approach to this problem (initiated by de Bruijn and Erdős [3] and continued by e.g. Vose [9], Tenenbaum and Yokota [8] and ChatGPT [2]) uses so-called *practical numbers*. A positive integer n is said to be practical if every positive integer smaller than n can be written as the sum of distinct divisors of n. To see why these are useful in the context of unit fractions, let a fraction $\frac{a}{b} \in (0,1)$ be given. We then choose a practical number $n > b$ and let m be such that $\frac{m}{n} \leq \frac{a}{b} < \frac{m+1}{n}$. As n is assumed to be practical, we can write both m and $an - bm < n$ as sums of divisors of n, say $m = d_1 + \cdots + d_r$ and $an - bm = e_1 + \cdots + e_s$. With this, we write

$$\begin{aligned}\frac{a}{b} &= \frac{m}{n} + \frac{an-bm}{bn} \\ &= \frac{d_1 + \cdots + d_r}{n} + \frac{e_1 + \cdots + e_s}{bn} \\ &= \frac{1}{n/d_1} + \cdots + \frac{1}{n/d_r} + \frac{1}{bn/e_1} + \cdots + \frac{1}{bn/e_s}\end{aligned}$$

as a sum of unit fractions. If n moreover has the property that r and s are small, then $\frac{a}{b}$ is the sum of a small number of unit fractions.

So, to reiterate: this strategy uses integers n for which *every* positive integer below n can be written as the sum of a *small* number of *divisors*. In the proof below we modify this and change all three words that were just emphasized.

For positive constants c and C, a positive integer n is said to be (c, C)-*semipractical*, or just semipractical for short, if every positive integer smaller than or equal to $e^{\frac{c \log n}{\log\log n}}$ can be written as the sum of at most C fractions $\frac{e}{f}$ with e and f positive integers and $e \mid n$.

To elaborate on how semipractical numbers differ from practical numbers: with semipractical numbers we no longer require every integer between 1 and n to be a sum, and we no longer require the summands to be integer divisors of n. On the other hand, the number of summands must now be uniformly bounded by this constant C.

2. Main theorem reduction

In this section we prove our main result, assuming the existence of certain semipractical numbers. More precisely, we will make use of the following result.

Lemma 1. *Let c be a constant with $0 < c < \frac{2\log 2}{3}$ and set $C := 10$. Then the integer $n := \operatorname{lcm}(1, 2, \dots, t)^2$ is (c, C)-semipractical for all sufficiently large t.*

We defer the proof of Lemma 1 to Section 3. Here we will simply assume that it holds and show how our main result follows from it.

Theorem 2. *Every fraction $\frac{a}{b} \in (0, 1)$ can be written as the sum of at most $\left(\frac{16}{\log 2} + o(1)\right) \log\log b$ distinct unit fractions.*

Proof. Let $\frac{a}{b} \in (0, 1)$ with b sufficiently large be given and define

$$n := \operatorname{lcm}\left(1, 2, \dots, \left\lceil \frac{\log b(\log\log b)^2}{2c} \right\rceil\right)^2.$$

By the prime number theorem we then find

$$n = e^{(\frac{1}{c} + o(1)) \log b (\log\log b)^2}, \tag{1}$$

and by semipracticality every positive integer at most

$$B := e^{\frac{c \log n}{\log\log n}} = e^{(1+o(1)) \log b \log\log b} \tag{2}$$

has a representation as the sum of at most C fractions whose numerators are divisors of n.

With the above in the back of our minds, the first thing we do is greedily remove $\frac{1}{x}$ from $\frac{a}{b}$ with x such that

$$\frac{1}{x} \le \frac{a}{b} < \frac{1}{x-1}.$$

The remainder can then be written as $\frac{a_1}{b_1}$ with $0 \le a_1 < a$ and $b_1 = bx$. We keep repeating this greedy process with $x = x_1, x_2, \dots, x_r$ where we write the remainder after i steps as $\frac{a_i}{b_i}$ with $b_i := bx_1x_2 \cdots x_i$ (and with $a_0 = a$ and $b_0 = b$). We continue for r steps, where r is the smallest integer for which we either have $a_r = 0$ or $b_r \ge n$. By induction one can then check that

$$\frac{a_i}{b_i} = \frac{a}{b} - \frac{1}{x_1} - \cdots - \frac{1}{x_i} \le \frac{1}{2^{2^{i-1}}} \tag{3}$$

for all $1 \le i \le r$. Combining the minimality of r with (3), we find

$$n > b_{r-1} \ge 2^{2^{r-2}},$$

which gives $r < \frac{\log\log n}{\log 2} + O(1)$. So if $a_r = 0$, then we have used at most

$$\frac{\log\log n}{\log 2} + O(1) = \left(\frac{1}{\log 2} + o(1)\right) \log\log b$$

unit fractions, which would prove the theorem. So without loss of generality we may assume that $a_r \geq 1$. We now have the following lemma.

Lemma 3. *For all $0 \leq i \leq r$ and all real numbers μ with $1 \leq \mu \leq b_i$, the integer b_i has a divisor in the interval $[\mu, b\mu]$.*

Proof. For $i = 0$ we can simply take the divisor b itself. So let $i \geq 1$ and assume by induction that the result holds for $i-1$. For all $\mu \leq b_{i-1}$ we can use the induction hypothesis, so we may further assume $b_{i-1} < \mu \leq b_i$. Now, by the greedy procedure,

$$x_i = \left\lceil \frac{b_{i-1}}{a_{i-1}} \right\rceil \leq b_{i-1}.$$

We can therefore apply the induction hypothesis with the real number $\frac{\mu}{x_i}$ to obtain a divisor d of b_{i-1} in the interval $\left[\frac{\mu}{x_i}, \frac{b\mu}{x_i}\right]$. Indeed, as

$$1 < \frac{\mu}{x_i} \leq \frac{b_i}{x_i} = b_{i-1},$$

the conditions for the induction hypothesis are satisfied. With this d, the integer dx_i then divides b_i and is contained in $[\mu, b\mu]$, as desired. □

Continuing the proof of our main result, we are going to write $a_r n$ as a sum of products $d_1 g_1 + \cdots + d_s g_s$ where $d_j \mid b_r$ and $1 \leq g_j \leq B$ for all $1 \leq j \leq s$. We do this iteratively: as long as

$$m_j := a_r n - d_1 g_1 - \cdots - d_j g_j > B,$$

we apply Lemma 3 with $\mu := \frac{m_j}{B}$ (which is allowed, because $1 \leq \mu < \frac{bn}{B} < n \leq b_r$), to obtain a divisor $d_{j+1} \in [\mu, b\mu]$, after which we choose $g_{j+1} := \left\lfloor \frac{m_j}{d_{j+1}} \right\rfloor$. As soon as $m_j = 0$ we stop, while for $1 \leq m_j \leq B$, we simply take $d_{j+1} = 1$ and $g_{j+1} = m_j$.

By the definitions of g_{j+1}, d_{j+1} and μ, we see

$$m_{j+1} < d_{j+1} \leq b\mu = \frac{bm_j}{B}$$

for all $0 \leq j \leq s-2$ (where $m_0 = a_r n$). In fact, as $m_s = 0$ by construction, the inequality $m_{j+1} < \frac{bm_j}{B}$ holds for all $0 \leq j \leq s-1$. That is, we started with $m_0 = a_r n < bn$ and the remaining integer gets reduced by a factor of at least $\frac{B}{b}$ every step. We therefore find

$$s \leq \frac{\log(bn)}{\log(B/b)} + 1 = \left(\frac{1}{c} + o(1)\right) \log\log b,$$

thanks to (1) and (2).

We now finally apply the semipracticality of n, to write $g_j = \sum_k \frac{e_{j,k}}{f_{j,k}}$ as a sum of at most C fractions for all $1 \le j \le s$, where the $e_{j,k}$ divide n. In conclusion we find

$$\begin{aligned}\frac{a}{b} &= \frac{1}{x_1} + \cdots + \frac{1}{x_r} + \frac{a_r}{b_r} \\ &= \frac{1}{x_1} + \cdots + \frac{1}{x_r} + \frac{d_1 g_1 + \cdots + d_s g_s}{b_r n} \\ &= \frac{1}{x_1} + \cdots + \frac{1}{x_r} + \frac{d_1 \sum_k e_{1,k}/f_{1,k}}{b_r n} + \cdots + \frac{d_s \sum_k e_{s,k}/f_{s,k}}{b_r n} \\ &= \frac{1}{x_1} + \cdots + \frac{1}{x_r} + \left(\sum_k \frac{1}{(b_r/d_1)(n/e_{1,k})f_{1,k}} \right) + \cdots + \left(\sum_k \frac{1}{(b_r/d_s)(n/e_{s,k})f_{s,k}} \right).\end{aligned}$$

In this final sum, the total number of summands is at most

$$r + Cs \le \left(\frac{1}{\log 2} + \frac{C}{c} + o(1) \right) \log\log b. \tag{4}$$

Since c can be chosen arbitrarily close to $\frac{2\log 2}{3}$, we deduce the claimed bound

$$\left(\frac{16}{\log 2} + o(1) \right) \log\log b$$

by taking $C = 10$. For a final note, even though these unit fractions are not necessarily distinct, Takenouchi's theorem [7, Sections II–III] guarantees that any sum of unit fractions (that equals a fraction in the interval $(0, 1)$) can be transformed into another one with the same number of summands, but where all unit fractions are distinct. □

3. Squares of lcms are semipractical

All that remains is to prove Lemma 1. To this end, we start with a lemma on short sums modulo primes.

Lemma 4. *For every odd prime p and for every set U of at least $p^{3/4}$ non-zero residue classes modulo p, every non-zero residue class $w \pmod p$ can be written as $u_1v_1 + u_2v_2$ with $u_1, u_2, v_1, v_2 \in U$.*

Proof. Let a non-zero residue class $w \pmod p$ be given. For an arbitrary pair (u_1, u_2) of residue classes modulo p (not necessarily contained in U^2), define

$$r(u_1, u_2) := \left| \{(v_1, v_2) \in U^2 : u_1v_1 + u_2v_2 = w\} \right|.$$

The goal is to prove $r(u_1, u_2) > 0$ for some pair $(u_1, u_2) \in U^2$. Now, on the one hand we have the equality

$$\sum r(u_1, u_2) = p|U|^2 \tag{5}$$

because every pair $(v_1, v_2) \in U^2$ gives rise to p pairs (u_1, u_2) with $u_1v_1 + u_2v_2 = w$. Indeed, u_1 can be chosen arbitrarily, after which u_2 is determined. On the other hand, we also claim that the upper bound

$$\sum r(u_1, u_2)^2 < |U|^4 + p|U|^2 \tag{6}$$

holds. To see this we note that $r(u_1, u_2)^2$ counts the number of ordered pairs $(v_1, v_2), (v_3, v_4) \in U^2$ that satisfy

$$u_1v_1 + u_2v_2 = u_1v_3 + u_2v_4 = w.$$

As in (5), every pair with $(v_1, v_2) = (v_3, v_4)$ contributes p choices of (u_1, u_2), with total contribution $p|U|^2$. On the other hand, the pairs $(v_1, v_2) \neq (v_3, v_4)$ (of which there are $|U|^4 - |U|^2 < |U|^4$) contribute at most one choice by the fact that we have two linear equations in two unknowns equal to the same non-zero value w.

We now combine (5) and (6) and apply Cauchy–Schwarz, in order to deduce

$$p^2|U|^4 = \left(\sum_{r(u_1,u_2)>0} 1 \cdot r(u_1, u_2)\right)^2 < |\{(u_1, u_2) : r(u_1, u_2) > 0\}| \left(|U|^4 + p|U|^2\right).$$

Rewriting this gives

$$\begin{aligned} |\{(u_1, u_2) : r(u_1, u_2) > 0\}| &> \frac{p^2|U|^4}{|U|^4 + p|U|^2} \\ &= p^2 - \frac{p^3}{|U|^2 + p} \\ &\geq p^2 - |U|^2, \end{aligned}$$

where the final inequality uses $|U| \geq p^{3/4}$. As the final expression $p^2 - |U|^2$ counts the number of pairs $(u_1, u_2) \notin U^2$, it follows that there must be a pair $(u_1, u_2) \in U^2$ with $r(u_1, u_2) > 0$, finishing the proof. □

We can now supply the final missing piece in the proof of Theorem 2.

Proof of Lemma 1. Let t be sufficiently large and let $m \leq B := e^{\frac{c \log n}{\log \log n}}$ be given. We then need to show that m can be written as the sum of at most 10 fractions $\frac{e}{f}$ with $e \mid n$. In fact, we will prove that there exists an m_0 (which is allowed to depend on c, but not on t or n) such that all odd integers m with $m_0 \leq m \leq B$ can be written as the sum of 9 such fractions whenever m is odd; for even $m > m_0$ one can then simply add $\frac{1}{1}$, while for all $m \leq m_0$ we note that m itself divides n as long as $t \geq m_0$, so that the sum with the single element m suffices.

So assume that m is an odd integer which is sufficiently large in terms of c, and define P to be the product of the first $k := 2 + \left\lceil \frac{3 \log m}{4 \log 2} \right\rceil$ primes. With U the set of divisors of P, we then have

$$|U| = 2^k \in [4m^{3/4}, 8m^{3/4}].$$

Now, we claim that the number of primes $p \leq m$ for which U occupies fewer than $p^{3/4}$ non-zero residue classes modulo p is $o\left(\frac{m}{\log^2 m}\right)$. To prove this claim, let Q be the set of such primes. We will then count the number of triples (u, v, p) with u and v distinct elements of U for which the difference between u and v is divisible by a prime $p \in Q$.

The difference between u and v is (in absolute value) bounded by P, so that every such difference (and there are fewer than $|U|^2$ of them) has at most $k < 2 \log |U|$ distinct prime factors. Hence, the number of these triples (u, v, p) is bounded by $2|U|^2 \log |U|$. On the other hand, for every $p \in Q$, U occupies fewer than

$$p^{3/4} + 1 \leq m^{3/4} + 1 < \frac{1}{2}|U|$$

residue classes modulo p, which means that there are at least $\frac{1}{2}|U|$ pairs of distinct $u, v \in U$ whose difference is divisible by p. Therefore, the number of triples is at least $\frac{1}{2}|U||Q|$, implying

$$|Q| < 4|U| \log|U| < 32m^{3/4} \log m = o\left(\frac{m}{\log^2 m}\right). \tag{7}$$

By Vinogradov's theorem on the odd Goldbach conjecture (see e.g. [5]), m can be written as the sum of three primes in at least $\frac{m}{20 \log^2 m}$ disjoint ways. Hence, by (7) it is possible to write $m = p_1 + p_2 + p_3$ as the sum of three primes for which U occupies at least $p_i^{3/4}$ non-zero residue classes modulo p_i for all $1 \leq i \leq 3$. So by Lemma 4 there exist

$$u_1^{(i)}, v_1^{(i)}, u_2^{(i)}, v_2^{(i)} \in U \qquad \text{with} \qquad u_1^{(i)} v_1^{(i)} + u_2^{(i)} v_2^{(i)} \equiv -1 \pmod{p_i}$$

for all $1 \leq i \leq 3$, say

$$u_1^{(i)} v_1^{(i)} + u_2^{(i)} v_2^{(i)} + 1 = p_i f_i.$$

We then finish the proof by writing

$$m = p_1 + p_2 + p_3 = \sum_{i=1}^{3} \left(\frac{u_1^{(i)} v_1^{(i)}}{f_i} + \frac{u_2^{(i)} v_2^{(i)}}{f_i} + \frac{1}{f_i} \right)$$

as a sum of 9 fractions, with $u_j^{(i)} v_j^{(i)} \mid P^2 \mid n$. Here, $P^2 \mid n$ follows from the fact that the number of primes dividing P is

$$k \leq \left(\frac{3c}{2 \log 2} + o(1) \right) \frac{t}{\log t} < \pi(t),$$

so every prime dividing P is smaller than t and therefore divides $\sqrt{n}$. □

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